

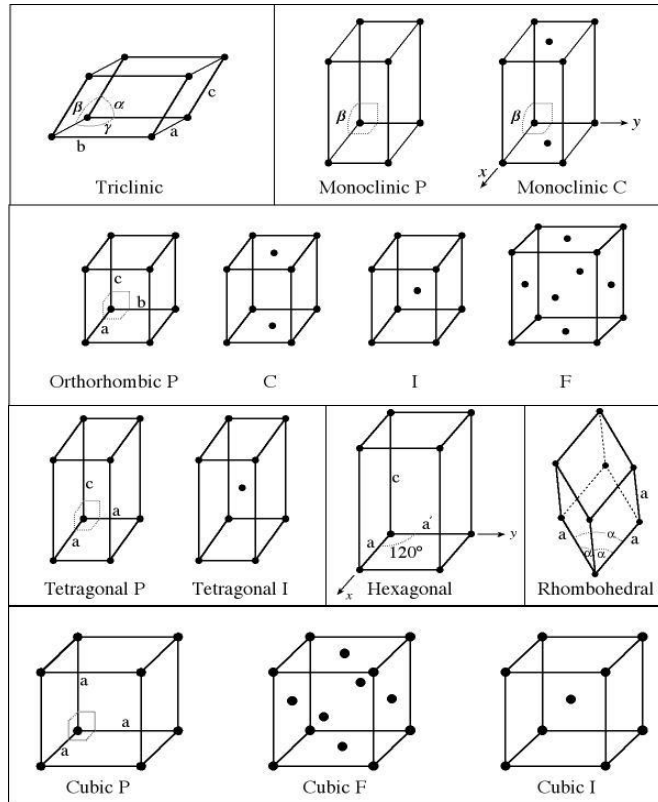
Equations, constants, tables and conversions for Exam 3

Constants:

$N_A = 6.02 \times 10^{23}$ atoms/mole; $h = 6.626 \times 10^{-34}$ J·s; $1 \text{ J} = 6.24 \times 10^{18} \text{ eV}$; $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

$e = 2.71828$; $\pi = 3.14159$

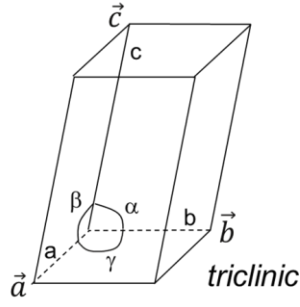
3D Bravais lattices:



Symmetry directions in point groups by crystal system:

Lattice	Primary	Secondary	Tertiary
Triclinic	None		
Monoclinic	unique (b or c)		
Orthorhombic	$[100]$	$[010]$	$[001]$
Tetragonal	$[001]$	$\{[100], [010]\}$	$\{[1\bar{1}0], [110]\}$
Hexagonal	$[001]$	$\{[100], [010], [\bar{1}\bar{1}0]\}$	$\{[1\bar{1}0], [120], [2\bar{1}0]\}$
Rhom. (hex.axis)	$[001]$	$\{[100], [010], [\bar{1}\bar{1}0]\}$	
Rhom. (rho)	$[111]$	$\{[1\bar{1}0], [01\bar{1}], [\bar{1}01]\}$	
Cubic	$\{[100], [010], [001]\}$	$\{[111], [1\bar{1}\bar{1}], [\bar{1}1\bar{1}], [\bar{1}\bar{1}1]\}$	$\{[1\bar{1}0], [110], [01\bar{1}], [011], [\bar{1}01], [101]\}$

Metric tensor:



$$\Gamma \equiv \begin{bmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{bmatrix}$$

$$\Gamma = \begin{bmatrix} a^2 & ab \cos \gamma & ac \cos \beta \\ ab \cos \gamma & b^2 & bc \cos \alpha \\ ac \cos \beta & bc \cos \alpha & c^2 \end{bmatrix}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \quad \vec{a} \cdot \vec{b} = [a_1 \quad a_2 \quad a_3][\Gamma] \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$|\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}} = \sqrt{[a_1 \quad a_2 \quad a_3][\Gamma] \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}}$$

$$L^2 = [\Delta x \quad \Delta y \quad \Delta z][\Gamma] \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

$$V = \vec{a} \cdot (\vec{b} \times \vec{c}) = abc \sqrt{1 - \cos^2 \alpha - \cos^2 \beta - \cos^2 \gamma + 2 \cos \alpha \cos \beta \cos \gamma}$$

Reciprocal lattice:

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V} = \frac{\vec{b} \times \vec{c}}{\vec{a} \cdot (\vec{b} \times \vec{c})} \quad \vec{b}^* = \frac{\vec{c} \times \vec{a}}{V} = \frac{\vec{c} \times \vec{a}}{\vec{a} \cdot (\vec{b} \times \vec{c})} \quad \vec{c}^* = \frac{\vec{a} \times \vec{b}}{V} = \frac{\vec{a} \times \vec{b}}{\vec{a} \cdot (\vec{b} \times \vec{c})}$$

$$\cos \alpha^* = \frac{\cos \beta \cos \gamma - \cos \alpha}{\sin \beta \sin \gamma} \quad \cos \beta^* = \frac{\cos \alpha \cos \gamma - \cos \beta}{\sin \alpha \sin \gamma} \quad \cos \gamma^* = \frac{\cos \alpha \cos \beta - \cos \gamma}{\sin \alpha \sin \beta}$$

$$\vec{k}_{hkl}^* = h\vec{a}^* + k\vec{b}^* + l\vec{c}^* \quad \vec{k}_{hkl}^* + \vec{k}_{h'k'l'}^* = \vec{k}_{(h+h')(k+k')(l+l')}^*$$

Reciprocal metric tensor:

$$\Gamma^* = \begin{bmatrix} \vec{a}^* \cdot \vec{a}^* & \vec{a}^* \cdot \vec{b}^* & \vec{a}^* \cdot \vec{c}^* \\ \vec{b}^* \cdot \vec{a}^* & \vec{b}^* \cdot \vec{b}^* & \vec{b}^* \cdot \vec{c}^* \\ \vec{c}^* \cdot \vec{a}^* & \vec{c}^* \cdot \vec{b}^* & \vec{c}^* \cdot \vec{c}^* \end{bmatrix} \quad \Gamma^* = \begin{bmatrix} a^{*2} & a^* b^* \cos \gamma^* & a^* c^* \cos \beta^* \\ a^* b^* \cos \gamma^* & b^{*2} & b^* c^* \cos \alpha^* \\ a^* c^* \cos \beta^* & b^* c^* \cos \alpha^* & c^{*2} \end{bmatrix}$$

$$|\vec{k}_{hkl}^*| = \frac{1}{d_{hkl}} \quad \boxed{\vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix}} \quad \cos \theta = \frac{\vec{k}_{h1k1l1}^* \cdot \vec{k}_{h2k2l2}^*}{|\vec{k}_{h1k1l1}^*| |\vec{k}_{h2k2l2}^*|}$$

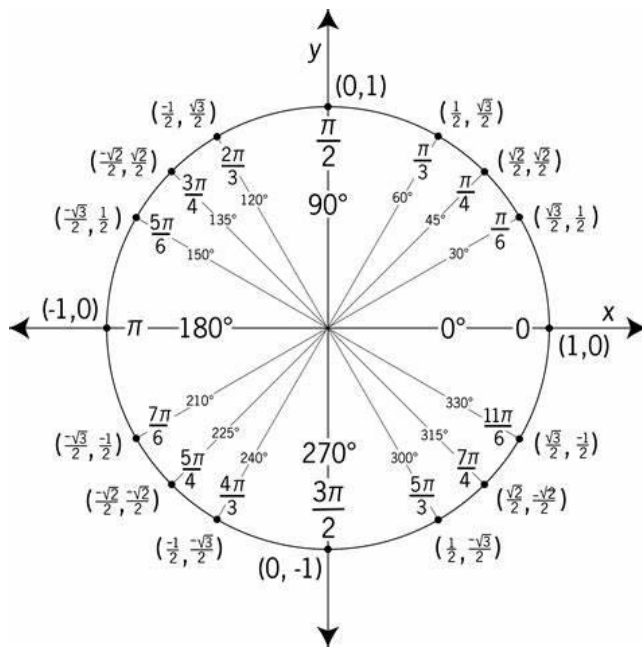
Photon energy:

$$E = h\nu = \frac{hc}{\lambda} \quad E/\text{eV} = \frac{1239.8}{\lambda/\text{nm}}$$

Bragg's law:

$$2d_{hkl} \sin \theta_{hkl} = \lambda \quad \frac{\vec{S} - \vec{S}_0}{\lambda} = \vec{k}_{hkl} = h\vec{a}^* + k\vec{b}^* + l\vec{c}^*$$

Unit Circle (of Life):



Bragg's law:

$$2d_{hkl} \sin \theta_{hkl} = \lambda \quad \frac{\vec{S} - \vec{S}_0}{\lambda} = \vec{k}_{hkl} = h\vec{a}^* + k\vec{b}^* + l\vec{c}^*$$

Structure factor:

$$S_{hkl} = \sum_{j=1}^N f_j e^{2i\pi(hx_j + ky_j + lz_j)} \quad I_{hkl} = |S_{hkl}|^2$$

Euler's formula: ($i = \sqrt{-1}$)

$$e^{i\theta} = \cos\theta + i \sin\theta \quad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Zone axis and zone law:

$$[uvw] = \vec{k}_{hkl} \times \vec{k}_{h'k'l'} \quad uh + vk + wl = 0$$

Intensity in an XRD experiment:

$$I_{hkl} = |S_{hkl}|^2 \cdot M_{hkl} \cdot Lp(\theta) \cdot e^{-B \left(\frac{\sin \theta}{\lambda} \right)^2} \quad Lp = \frac{1 + \cos^2 2\theta}{\sin^2 \theta \cos \theta} \quad f(s) = Z - 41.78214 \cdot s^2 \cdot \sum_{i=1}^4 a_i e^{-b_i s^2}$$

M_{hkl} :

Cubic	hkl 48	hhl 24	OkI 24	Okk 12	hhh 8	00I 6	
Hexagonal/rhombic	hkl 24	hhl 12	OkI 12	hk0 12	hh0 6	Ok0 6	00I 2
Tetragonal	hkl 16	hhl 8	OkI 8	hk0 8	hh0 4	Ok0 4	00I 2
Orthorhombic	hkl 8	OkI 4	h0I 4	hk0 4	h00 2	Ok0 2	00I 2
Monoclinic	hkl 4	h0I 2	00I 2				
Triclinic	hkl 2						

Systematic absences:

Table 4.1. Systematic absences for centred unit cells.

Centring	Points equivalent to 0,0,0	Condition for observed intensity	Fraction of observed data
<i>P</i>	none	none	1
<i>A</i>	0, $\frac{1}{2}$, $\frac{1}{2}$	$k + l = 2n$	$\frac{1}{2}$
<i>B</i>	$\frac{1}{2}$, 0, $\frac{1}{2}$	$h + l = 2n$	$\frac{1}{2}$
<i>C</i>	$\frac{1}{2}$, $\frac{1}{2}$, 0	$h + k = 2n$	$\frac{1}{2}$
<i>I</i>	$\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$	$h + k + l = 2n$	$\frac{1}{2}$
<i>F</i>	0, $\frac{1}{2}$, $\frac{1}{2}$ and $\frac{1}{2}$, 0, $\frac{1}{2}$ and $\frac{1}{2}$, $\frac{1}{2}$, 0	h, k, l all even or all odd	$\frac{1}{4}$
<i>R</i>	$\frac{1}{3}$, $\frac{2}{3}$, $\frac{2}{3}$ and $\frac{2}{3}$, $\frac{1}{3}$, $\frac{1}{3}$	$-h + k + l = 3n$	$\frac{1}{3}$

Table 4.2. Systematic absences for glide planes.

Normal to glide plane	Reflections affected	Glide plane symbol	Translation vector	Condition for observed intensity
<i>a</i> -axis [100]	<i>okl</i>	<i>b</i>	b /2	$k = 2n$
		<i>c</i>	c /2	$l = 2n$
		<i>n</i>	(b+c)/2	$k + l = 2n$
		<i>d</i>	(b+c)/4	$k + l = 4n$
<i>b</i> -axis [010]	<i>h0l</i>	<i>a</i>	a /2	$h = 2n$
		<i>c</i>	c /2	$l = 2n$
		<i>n</i>	(a+c)/2	$h + l = 2n$
		<i>d</i>	(a+c)/4	$h + l = 4n$
<i>c</i> -axis [001]	<i>hk0</i>	<i>a</i>	a /2	$h = 2n$
		<i>b</i>	b /2	$k = 2n$
		<i>n</i>	(a+b)/2	$h + k = 2n$
		<i>d</i>	(a+b)/4	$h + k = 4n$
[110]	<i>hhl</i>	<i>c</i>	c /2	$l = 2n$
		<i>d</i>	(a+b+c)/4	$2h + l = 4n$

Table 4.3. Systematic absences for screw axes: conditions for observed intensity.

Parallel to	Reflections affected	Condition for $2_1, 4_2$ or 6_3	Condition for $3_1, 3_2, 6_2$ or 6_4	Condition for 4_1 or 4_3	Condition for 6_1 or 6_5
<i>a</i>	<i>h</i> 00	$h = 2n$	$h = 3n$	$h = 4n$	$h = 6n$
<i>b</i>	0 <i>k</i> 0	$k = 2n$	$k = 3n$	$k = 4n$	$k = 6n$
<i>c</i>	00 <i>l</i>	$l = 2n$	$l = 3n$	$l = 4n$	$l = 6n$